

SAMBHAV VERMAN

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PROFILE

- Over 15 years of experience in the Information Technology industry.
- 5 years of work experience in the Mac OSX GPU video graphics driver team at AMD.
- 4 years of work experience as a scientist in the avionics industry (DRDO, India).
- Strong knowledge in OS fundamentals, cryptography, embedded systems, data structures and algorithms.
- Strong debugging and bug triaging skills for single and multithreaded applications.
- Knowledge and experience in FIPS-140, automotive ISO 26262, military DO-160 and DO-178B standards.
- Experience in delivering under Agile and Waterfall project management methodologies.

WORK EXPERIENCE

General Electric, Markham, ON, Canada. (Senior Embedded Developer/Contractor)	(Aug 2024 – Present)
Kinaxis Inc, Toronto, ON, Canada. (Software Developer)	(April 2022 – May 2024)
Cryptomill Cybersecurity Solutions, Toronto, ON, Canada (Sr. Software Engineer)	(June 2021 – Mar 2022)
Qualcomm, Markham, ON, Canada. (Sr. Software Engineer)	(Mar 2020 – June 2021)
AMD, Markham, ON, Canada. (Sr. Software Engineer)	(Oct 2016 – Feb 2020)
AMD, Sunnyvale, CA, U.S. (Sr. Software Engineer)	(Jan 2015 – Sep 2016)
CISCO, Milpitas, CA, U.S. (Software Engineer Contractor)	(July 2014 – Jan 2015)
Defense Research and Development Organization of India, India. (Scientist 'C')	(May 2008 - July 2012)

TECHNICAL SKILLS

Operating Systems	- Windows, Linux and Mac OS X.
Programming Languages	- C, C++, Objective C, C#, Python and AT&T Assembly.
Miscellaneous	- Bash, Perforce, Git VCS, JIRA, Xcode, MS Visual Studio and Eclipse IDE, Machine Learning, MongoDB and MYSQL.

WORK EXPERIENCE DETAILS

- I. **General Electric (GE), (Contract via Alten Canada)** **Aug 2024 – Present**
Senior Embedded Developer
Firmware design and development for GE Universal Relay (UR) products (C++) :
 1. Design and develop microservices in the firmware (modern C++ 17).
 2. Build, run and test the firmware on Yocto based RT-linux running on Xilinx FPGA based SoCs.
 3. Execute in a fast paced Scaled Agile framework (SAFe) environment.**Technological environment: Linux, C++ 11/17, Redis DB, Docker, SAFe Agile, Microservices, Xilinx FPGA, Xen hypervisor, RT-Linux, JIRA and Git.**

- II. **Kinaxis Inc,** **April 2022 – May 2024**
Software Developer
Development & Support for Rapid Response : A Supply Chain Analytics software:
 1. Fix customer issues in the core analytics code base (C++ 17).
 2. Enhance existing features to address customer requests.
 3. Write Unit tests using NUnit and Google Test frameworks (C# and C++).
 4. Execute in a fast paced Agile based environment.**Technological environment: Windows, C++17, C#, Visual Studio, DS Jenkins CI/CD, Agile and Git.**

III. Cryptomill Cybersecurity Solutions

June 2021 - Mar 2022

Senior Software Engineer

Back-end and infrastructure development for Circles of Trust – an EndPoint encryption technology based on a Data Centric Security model :

1. Added portability and IPC support for Linux platform. Designed a UI app using QT for decryption on Linux.
2. Added an update manager on Windows platform for pulling software updates from the cloud.
3. Created a command line tool which dumps diagnostic logs into files and uploads to a file handling database (MongoDB gridFS). This would facilitate debugging for remote machines.

Technological environment: Windows, C++14, Visual Studio, Git, SVN and PowerShell.

IV. Qualcomm

Mar 2020 – June 2021

Senior Software Engineer

Development and support of automotive display drivers on QNX OS :

1. Ported and verified several features from old to new generation automotive display drivers.
2. Wrote a detailed display architecture document for the QNX display drivers.
3. Developed and automated code coverage collection for display drivers using Parasoft tools for compliance to ISO26262 standard.

Technological environment: QNX RTOS, ISO26262, Parasoft tools, Snapdragon SoCs, C language, JIRA and Git.

V. AMD Inc

Jan 2015 - Feb 2020

Senior Software Engineer

Development and support of video graphics driver for external graphics cards (GPUs) on Mac OSX platform :

1. Directly responsible for maintaining video drivers and implement new features for next generation AMD GPUs.
2. Provided end to end debugging support to the customer (Apple) for video graphics driver related issues. Fixed a number of high priority issues meeting the deliverable timeline. On numerous occasions, this involved communicating to the client engineers on-site (Apple, Cupertino campus) and effectively working out a solution by sharing debug drivers and collecting/analyzing debug logs.
3. Worked independently and at times with the team to resolve complex issues related to bugs in the video driver stack, firmware and sometimes in the hardware as well. Typical issues would involve gathering and analyzing huge data logs and narrowing down the root cause.
4. Optimize the performance and stability of the drivers and firmware for video encoder and decoder. Added several features in the driver to help ease the task of isolating the bugs. Improved the debugging logs, wrote scripts to automate tests. Carried out regressions in the driver versions numerous times to isolate issues with change-lists (user submissions).
5. Developed a new feature that helped prioritize job submissions to the hardware improving the overall throughput of the video driver for certain tasks. This would also prevent GPU deadlock in certain scenarios. Independently, added support for new Powerplay library bring-up (Powerplay library is software component which turns GPU power on/off).

Technological environment: Mac OS, C++, Objective-C, AMD GPUs, video graphics, device drivers, Xcode and Perforce.

VI. Cisco Systems Inc.

July 2014 – Jan 2015

Software Engineer (Contract via Experis Inc)

Implementation and testing of Cisco's TPM aka Trust Anchor Module (TAM) – a technology for encryption in hardware :

1. Developed a true random number generator (TRNG) used in cryptographic operations on the firmware of the ACT chip (CISCO's Anti Counterfeit Technology). The TRNG was realized on a SoC

- SmartFusion2 platform from Microsemi. Also developed the health tests (as per FIPS recommendation) for validation. [VHDL and C]
2. Integration and porting of the old ACT firmware to the new SoC SmartFusion2 platform.
 3. Porting of cryptographic encryption and hash functions to the FPGA platform. Validation and testing for conformance to FIPS-140.

VII. C.A. Technologies

Dec 2012 – Dec 2013

Research Assistant

Development of a wrapper application on iOS platform to implement security policies for iOS apps using a security application (Objective C) :

Description- The security application injects a dynamic library into a working iOS app and carries out security checks against the security policies stored on a remote server.

Role- Developed the security backend that formulates security policies for trapped method calls.

Developed the shared memory model (using POSIX multithreading techniques) to trap method calls to various resources within the iOS environment coming from different apps. The security backend then logs and transmits the data to a remote server which regulates and generates security policies for the device.

VIII. Defense Research and Development Organization of India

Dec 2008 – Apr 2012

Scientist 'C'

1. Development of firmware and application software for an FPGA based avionics package used in various aircraft systems for command reception and transmission. (DRDO, Jan 2010–Apr 2012)

Description - AIU is an FPGA based avionics package used to transmit and receive digital and analog data through various other packages such as telemetry system and sensors present on the on-board system of an aircraft.

Role – Developed the firmware and application software for the avionics package [using C and VHDL].

Developed the test application software which carries out checks for all the resource on the FPGA based avionics package [C and C++].

2. Development of application software on VxWorks RTOS for the aircraft communication system.

Description - The aircraft communication system consists of a VxWorks based C application running on a ruggedized industrial PC and guides an airborne system.

Role- Developed the application software for the communication system which included initializing and interfacing various communication components (Ethernet and ARINC devices) and data acquisition devices [using C].

Technological environment: Linux, Vxworks RTOS, Xilinx FPGA SoCs, DO-178B, DO-160, C, C++ and VHDL.

EDUCATION

M.S. in Electrical Engineering

Aug.2012 - Dec 2013

State University of New York at Stony Brook.

Bachelor of Technology in Electronics and Communication Engineering

Aug 2003 - Nov 2007

Uttar Pradesh Technical University, India

Languages

English, Hindi